NAMA PEMILIK PROYEK : **PT PERTAMINA EP**

NAMA PROYEK : **JASA PEMBANGUNAN FASILITAS PRODUKSI STASIUN PENGUMPUL (SP) TANJUNG MIRING BARAT & KARANGAN TIMUR**

LOKASI : **SP TANJUNG MIRING BARAT & KARANGAN TIMUR - FIELD LIMAU**

NO. KONTRAK : **SA02035383B**

REV.	TGL.	DESKRIPSI	DISIAPKAN	DIPERIKSA	DISETUJUI	DIPERIKSA	DISETUJUI
A	18/12/2025	Issued For Review	AP	AS	RTN		
			PT. PROSYMPAC OIL & GAS			PT. PERTAMINA EP	

TABEL REVISI

Rev.	Tanggal	Deskripsi

COMMENT RESPONSE SHEET

No.	Page	Company's Comment	Contractor's Response	Remarks

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1. GENERAL

1.1 Latar Belakang

Struktur Tanjung Miring Barat (TMB) terletak di Kabupaten Muara Enim dan Kota Prabumulih, sekitar 20 kilometer di sebelah tenggara Kota Prabumulih, dengan luas daerah sekitar 8 km². Struktur Karangan (KRG) terletak di Kabupaten Muara Enim dan Kota Prabumulih, sekitar 15 kilometer di sebelah selatan Kota Prabumulih, dengan luas daerah sekitar 45 km².

ID	Easting (Meter)	Northing (Meter)	Elevation (Meter)	Feature Code
PN-1531	424962,745	967376,210	10,414	
PM-8593	425000,093	907528,777	19,046	

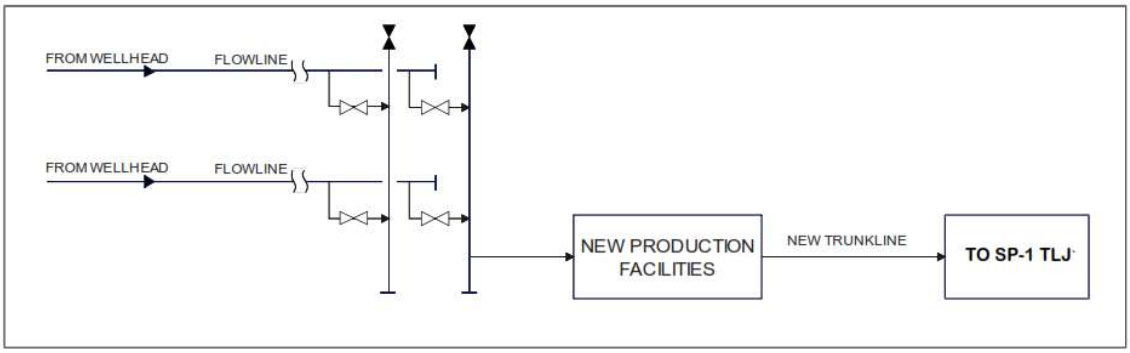
Gambar 1. Lokasi Rencana SP Tanjung Miring Barat - Karangan Timur

Lapangan Tanjung Miring Barat telah diproduksi sejak tahun 1938 dan terus beroperasi hingga tahun 2003 dengan produksi terakhir dari sumur TMB-11. Sejak dinonaktifkannya lapangan ini, fasilitas produksi yang terdiri dari flowline, stasiun pengumpul, serta jalur pengiriman pipa minyak ke custody transfer (PPP Prabumulih) sudah tidak layak operasi. Apabila akan dilakukan pengembangan lapangan, diperlukan pembangunan fasilitas produksi baru yang dapat menampung produksi hidrokarbon dari lapangan Tanjung Miring Baral serta memprosesnya agar sesuai dengan spesifikasi yang dipersyaratkan di titik serah. Saat ini status SP TMB adalah idle atau tidak beroperasi, sehingga produksi sumuran menggunakan Tank On-Site (TOS) dan dikirim secara gross ke SP-II TMT menggunakan trucking. Dikarenakan tidak adanya line gas ke sumur-sumur struktur TMB, supply Listrik untuk pumping unit menggunakan solar (diesel engine). Lapangan Karangan Timur hanya diproduksikan dari 2 sumur yaitu KRG-9 dan KRG-10 beroperasi mulai tahun 1984, dimana kondisi eksisting jalur pengiriman minyak dari sumur menggunakan trucking. Apabila akan dilakukan pengembangan lapangan, diperlukan pembangunan fasilitas produksi baru yang dapat

menampung produksi hidrokarbon dari lapangan Tanjung Miring Barat serta memprosesnya agar sesuai dengan spesifikasi yang dipersyaratkan di titik serah.

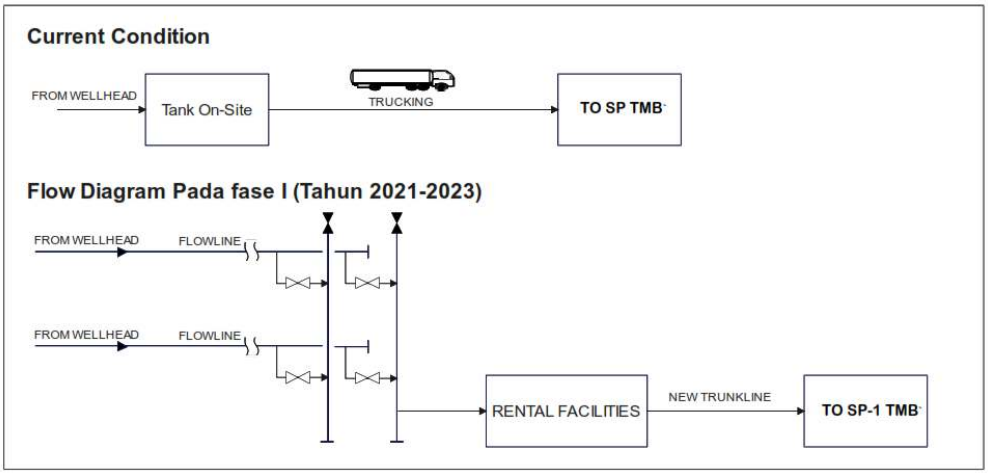
Dalam rencana jangka panjang sebagai upaya peningkatan produksi, PT Pertamina EP berencana akan mengembangkan Lapangan Tanjung Miring Barat dan Karangan Timur melalui pengembangan fasilitas produksi dimana rencana kerja pengembangan fasilitas akan dibagi ke dalam dua fase.

Pada fase pertama, produksi dari pengembangan awal TMB -KRG Timur akan dikirimkan menuju fasilitas sewa Early Production Facility (EPF) untuk selanjutnya dikirimkan menuju SP-1 TLJ melalui trunkline baru. Pada fase ini lingkup fasilitas produksi yang akan dibangun adalah flow/ine dari sumuran, sewa EPF dan trunkline dari EPF sampai dengan SP-1 TLJ.



Gambar 2. Rencana Pola Operasi Fase

Pada fase kedua, akan dilakukan pengembangan tambahan dari sumur-sumur TMB-KRG Timur. Pada fase ini, lingkup fasilitas produksi yang akan dibangun meliputi pembangunan flowline dari sumuran pengembangan lanjutan serta peningkatan fasilitas SP TMB untuk menggantikan sewa EPF.



Gambar 3. Rencana Pola Operasi Fase

1.2 Tujuan

Dokumen ini menjelaskan dipersiapkan sebagai Mechanical Datasheet KOD Pump (P-006 A/B) sebagai bagian dari desain yang harus diikuti oieh KONTRAKTOR dan VENDOR untuk proyek Pembangunan Fasilitas Stasiun Pengumpul Lapangan Tanjung Miring Barat- Karangan Timur, Field Prabumulih.

1.3 Ruang Lingkup

Mechanical Datasheet ini meliputi ketentuan minimal dalam mengatur desain. material, fabrikasi, pemeriksaan, pengujian, persiapan, pengecatan dan persiapan untuk pengiriman KOD Pump (P-006 A/B) untuk dipasang di proyek Pembangunan Fasilitas Stasiun Pengumpul Lapangan Tanjung Miring Barat — Karangan Timur, Field Prabumulih.

1.4 Istilah

Dokumen ini berisi istilah-istilah yang memiliki definisi tercantum sebagai berikut:

Perusahaan	Perusahaan yang dimaksud adalah PT PERTAMINA HULU ROKAN selaku Pemilik dan Pengguna Fasilitas akhir.
Kontraktor	Pihak yang dipilih oleh PERUSAHAAN sebagai pelaksana pekerjaan pengadaan dan konstruksi yang akan bertanggung jawab kepada PERUSAHAAN.
Vendor	Pihak lain yang akan menyediakan sesuatu untuk kebutuhan proyek, selain yang disediakan oleh PERUSAHAAN dan KONTRAKTOR, disebut VENDOR. Tanggung jawabnya langsung kepada KONTRAKTOR baik terhadap PO maupun bentuk perjanjian lain antara VENDOR dan KONTRAKTOR.

SKK MIGAS	Suatu badan dari Pemerintah Indonesia yang bertanggung jawab terhadap proses persetujuan dan izin untuk semua fasilitas di lingkungan minyak dan gas.
Perencana	Pihak yang dipilih oleh PERUSAHAAN sebagai pelaksana pekerjaan Jasa FEED Engineering Services. Dalam hal ini disebut PERENCANA, yang akan bertanggung jawab kepada PERUSAHAAN sebagaimana dirinci dalam kontrak, scope of work, atau work order.
Inspektor	Perwakilan dari PERUSAHAAN dan/atau seseorang yang ditunjuk oleh PERUSAHAAN, yang bertanggung jawab sebagai INSPEKTOR atas pekerjaan yang dilakukan oleh KONTRAKTOR.SINGKATAN

2. KODE, STANDAR, DAN REFERENSI

DOKUMEN PROYEK

1. TMBP-SP001-ME-DBS-D101 — Mechanical Design Basis
2. TMBP-SP001-PR-DST-D108 —Process Datasheet – KOD Pump P-006 A/B
3. TMBP-SP001-PR-UFD-D103_UTILITY_FLOW_DIAGRAM_FOR_FLARE_SYSTEM
4. TMBP-SP001-ME-SPE-D103 — Specification – API Centrifugal Pump

3. Daftar Singkatan

Beberapa singkatan yang paling umum digunakan tercantum dalam daftar berikut:

BOPD	*Barrel Oil Per Day*
BPD	*Barrel Per Day*
BWPD	*Barrel Water Per Day*
EOR	*Enhanced Oil Recovery*
FEED	*Front End Engineering Design*
KRG	*Karangan*
MMSCFD	*Million Standard Cubic Feet per Day*
NPS	*Nominal Pipe Size*
PB	*Production Battery*
PHR	*Pertamina Hulu Rokan*

PT	*Perseroan Terbatas*
SP	*Stasiun Pengumpul*
SCH	*Schedule*
TMB	*Tanjung Miring Barat*
TMT	*Tanjung Miring Timur*
WK	*Wilayah Kerja*

GENERAL

APPLICABLE TO:	PROPOSAL	APPLICABLE NATIONAL / INTERNATIONAL STANDARD :	API-610 (Note 3)	Rev
CLIENT	Pertamina EP Regional 1 Zona 4	UNIT		
SITE	Fasilitas Produksi Stasiun Pengumpul (SP)	SERVICE	Closed Drain Pump	
NO. REQ	2x100% (Note 12)	PUMP SIZE	*	TYPE: OH2 — No. STAGES: 1 (One)
MANUFACTURER	*	MODEL	*	SERIAL NO. *
EQUIPMENT NUMBER	P-007 A/B			

LIQUID CHARACTERISTICS

	Units						
LIQUID TYPE OR NAME :		Mixed Oil & Water					
		Rated	Normal	MAX	MIN		
SERVICE :		Process Liquid					
VAPOR PRESSURE :	(psi a)	2.75					
SG @ Pump. Temp (3.1.53) :		0.88					
SPECIFIC HEAT :	(Btu/lb)						
VISCOSITY @ Pump. Temp. :	(cP)	12.66					

OPERATING CONDITIONS (6.1.3)

	Units	Rated	Normal	MAX	MIN	Alt Condition 1 {Name 3}	Alt Condition 2 {Name 4}
NPSHA Datum:		C.L. Impeller					
PUMPING TEMPERATURE :	(ºF)	65.00					
FLOW (Note 11) :	(BPD)	550					
DISCHARGE PRESSURE (6.3.2) :	(psig)	30.0					
SUCTION PRESSURE :	(psig)	1.65					
DIFFERENTIAL PRESSURE :	(psi)	28.35					
DIFFERENTIAL HEAD :	(ft)	74.25					
NPSHA (Note 1) :	(ft)	35.81					
HYDRAULIC POWER:	(hp)	0.30					

SERVICE (CONTINUOUS/INTERMITTENT) :	INTERMITTENT	INSTALLATION LOCATION :	OUTDOOR
• IF INTERMITTENT, NO. OF STARTS :		IF INDOOR, TEMPERATURE: MAX: ºC	MIN: ºC
PUMPS OPERATE IN:	ALONE	ELECTRIC AREA CLASSIFICATION (6.1.29):	HAZARDOUS
CORROSION DUE TO : (6.12.1.9)	N/A	CLASS	ZONE 2
EROSION DUE TO : (6.12.1.9)		GROUP IIA	LIQUID CLASS: FLAMMABLE
H2S CONCENTRATION (ppm) : (6.12.1.13)	N/A	TEMP CLASS T3	
CHLORIDE CONCENTRATION	N/A	PUMP VALVE START POSITION :	Open Valve (Fully Loaded)
PARTICULATE SIZE (DIA IN MICRONS)	Note 1		
PARTICULATE CONCENTRATION (ppm)	N/A		
CORROSION ALLOWANCE (6.3.10) :			
DESIGN NOTES:	Tropicalization Required		

PERFORMANCE

PROPOSAL CURVE NO.	*	RPM	*	
IMPELLER DIA.: RATED	*	MAX. *	MIN. *	(in)
RATED POWER :	* HP	EFFICIENCY	*	%
RATED CURVE BEP FLOW (at rated impeller diameter) (6.1.16) :			*	(bpd)
MIN FLOW : THERMAL	* (bpd)	STABLE	*	(bpd)
PREFERRED OPERATING REGION (6.1.16)	*	to	*	(bpd)
ALLOWABLE OPERATING REGION (6.9.4.1)	*	to	*	(bpd)
MAX HEAD @ RATED IMPELLER			*	(ft)
MAX POWER @ RATED IMPELLER			*	(HP)
NPSH3 AT RATED FLOW (6.1.9) :			*	(ft)
CENTERLINE OF PUMP TO NPSHA DATUM			*	(ft)
NPSH MARGIN AT RATED FLOW (6.1.10) :			*	(ft)
SPECIFIC SPEED (6.1.17)		m3/h, rpm, m	*	(m3/h, rpm, m)
SUCTION SPECIFIC SPEED LIMIT (6.1.11)		m3/h, rpm, m	*	
SUCTION SPECIFIC SPEED (6.1.11)		m3/h, rpm, m	*	(m3/h, rpm, m)
MAX. ALLOWABLE SOUND PRESSURE LEVEL (6.1.19)			85 @1 m	(dBA)
ESTIMATED MAX. SOUND PRESSURE LEVEL			*	(dBA)
MAX. ALLOWABLE SOUND POWER LEVEL (6.1.19)				(dB)
ESTIMATED MAX SOUND POWER LEVEL			*	(dB)

DRIVER (7.1) (Note 4)

DRIVER TYPE	Electric Motor
GEAR	NO
VARIABLE SPEED REQUIRED	NO
SOURCE OF VARIABLE SPEED	N/A
MANUFACTURER	*
NAMEPLATE POWER : HP	*
NOMINAL RPM	*
RATED LOAD RPM	*
FRAME	*
ORIENTATION	HORIZONTAL
LUBRICATION	*
BEARING TYPE	Anti Friction
RADIAL	*
THRUST	*
STARTING METHOD	DOL
DRIVER DATASHEET NUMBER	TMBP-SP001-EL-DST-D102
DRIVE DESIGN STANDARD	IEC

CONSTRUCTION

API PUMP TYPE (Table 1): OH2

NOZZLE CONNECTIONS (6.4.2):

	Size	Facing	Rating	Position
SUCTION	3" (*)	RF	300	END
DISCHARGE	2" (*)	RF	300	TOP

PRESSURE CASING AUX. CONNECTIONS: (6.4.3)

	No.	Size	Type	Facing	Rating	Posn.
BALANCE/LEAK OFF						
DRAIN **(Note 6)**		(*)		RF	(*)	
VENT **(Note 6)**		(*)		RF	(*)	
PRESSURE GAGE						
TEMP GAGE						
WARM-UP LINE						

DRAIN VALVE SUPPLIED BY (7.6.2.5)	SUPPLIER
DRAINS MANIFOLDED (7.6.2.5)	NO
VENT VALVE SUPPLIED BY	SUPPLIER
VENTS MANIFOLDED	NO
THREADED CONS FOR PIPELINE SERVICE & < 55°C (6.4.3.2)	
SPECIAL FITTINGS FOR TRANSITIONING (6.4.3.3)	
CYLINDRICAL THREADS REQUIRED (6.4.3.2)	
MACHINED AND STUDDED AUX CONNECTIONS (6.4.3.6)	
ROUTE DRAIN TO SKID EDGE	YES

CASING MOUNTING (6.3.14, 9.3.8.3): CENTERLINE

CASING TYPE: SINGLE VOLUTE

OH3 BACK-PULLOUT LIFTING DEVICE REQD. (9.1.2.6): N/A

CASING PRESSURE RATING:

MAWP (6.3.6) : * psig @ * (ºF)

HYDROTEST (8.3.2.2) : * psig @ AMB (ºF)

WETTING AGENT REQUIRED FOR HYDROTEST (8.3.2.7) :	*
HYDROTEST OH PUMP AS ASSEMBLY (8.3.2.14) :	*
SUCT'N PRESS. REGIONS DESIGNED FOR MAWP (6.3.8) :	YES

ROTOR:

SHAFT FLEXIBILITY INDEX (SFI) (9.1.1.3)	*
FIRST CRITICAL SPEED WET (MULTISTAGE PUMPS ONLY)	N / A
COMPONENT BALANCE TO ISO 1940-1, G1 (6.9.3.4) :	YES
SHRINK-FIT-LIMITED MOVEMENT IMPELLERS (9.2.2.3)	*
ROTATION (VIEWED FROM COUPLING END) :	CW
IMPELLERS INDIVIDUALLY SECURED (6.6.3) :	YES

COUPLING: (7.2)

MANUFACTURER	*
MODEL	Flexible Disc Type with Spacer (*)
RATING (kW / 100 RPM)	*
SPACER LENGTH (7.2.2.d)	* mm
SERVICE FACTOR (7.2.3)	1.5
RIGID-TYPE	NO
COUPLING WITH HYDRAULIC FIT (7.2.9)	*
COUPLING WITH PROPRIETARY CLAMPING DEVICE (7.2.10)	*
COUPLING IN COMPLIANCE WITH (7.2.4)(7.2.2.f)	API 671

GUARDS (7.3)

COUPLING AND SHAFT GUARD STANDARD (7.3)	*
IGNITION HAZARD ASSESSMENT PER EN 13463-1 (7.3.2.2; 7.3.3.4)	*
COUPLING GUARD MATERIAL (7.3.2.1; 7.3.3.3) :	Non-sparking
SHAFT GUARD MATERIAL (7.3.2.1; 7.3.3.3) :	*
SPARK RESISTANT MATERIAL REQUIRED (7.3.2.1) :	YES

MATERIAL (6.12.1.1) (Note 8)

ANNEX H CLASS	S-6	
MINIMUM DESIGN METAL TEMPERATURE (6.12.4.1)	0	°C
MAXIMUM ALLOWABLE TEMPERATURE (3.1.19)		°C
REDUCED-HARDNESS MATERIALS REQ'D (6.12.1.14)		
APPLICABLE HARDNESS STANDARD (6.12.1.14)		
BARREL :		
CASE :	CS	
DIFFUSERS		
IMPELLER :	12% Hardened	
IMPELLER WEAR RING :	12% Hardened	
CASE WEAR RING :	12% Hardened	
SHAFT:	4140 Alloy Steel	
BOWL (IF VS-TYPE)		

BEARINGS AND BEARING HOUSINGS (6.10.1)

BEARING (TYPE / NUMBER):

RADIAL	*	/	*
THRUST	*	/	*

REVIEW AND APPROVE THRUST BEARING SIZE : (9.2.5.2.6)	
LUBRICATION (6.11.3) (9.3.12.4) : **(Note 11, 14)**	FLOOD
PRESSURE LUBE SYSTEM TO API-614, CHAPTER :	*
API 614 DATASHEETS ATTACHED (9.2.6.4)	*
PRESSURIZED LUBE OIL SYSTEM MOUNTED ON BASEPLATE:	*
LOCATION OF PRESSURIZED LUBE OIL SYSTEM:	*
INTERCONNECTING PIPING PROVIDED BY	*
OIL VISC. ISO GRADE	*
VENT-TO-HOUSING CONSTANT LEVEL OILER (6.10.2.4) :	REQUIRED
OIL MIST PROVISIONS (6.11.3)	N/A
GREASE LUBRICATION (6.11.4)	N/A

BASEPLATE

API BASEPLATE NUMBER (ANNEX D) :	*
IF NON-STD BASEPLATE DIMENSIONS (LxW) (	
BASEPLATE CONSTRUCTION (7.4)	*
BASEPLATE DRAINAGE (7.4.1)	*
MOUNTING :	GROUTED
NON-GROUT CONSTRUCTION (7.4.1.e):	
SUPPLIED WITH : ● GROUT AND VENT HOLES	YES
● DRAIN CONNECTION	YES
DEMONSTRATE BASEPLATE PAD FLATNESS (7.4.9)	*
PROVIDE SPACER PLATE UNDER ALL EQUIPMENT FEET (7.4.10)	
BOLT OH 3/4/5 PUMP TO PAD / FOUNDATION :	N / A
PROVIDE SOLEPLATE FOR OH 3/4/5 PUMPS (9.1.2) :	N / A

REMARKS:

DISCHARGE CIRCUIT PIPING DESIGN:

DESIGN PRESSURE _____ psig

DESIGN TEMPERATURE _____ ºF

Rev

INSTRUMENTATION

INSTRUMENTATION PER API-670 (7.5.2)	N / A
ACCELEROMETERS (7.5.2.1)	
NUMBER OF ACCELEROMETERS	N/A
MOUNTING LOCATION:	
PROVISION FOR MTG ONLY (6.10.2.13)	
FLAT SURFACE REQUIRED (6.10.2.14)	
VIBRATION PROBES (7.5.2.2)	
VIBRATION PROBES REQUIRED (7.5.2.2)	N / A
NUMBER PER RADIAL BEARING	
NUMBER PER AXIAL BEARING	
THREADED PROVISION FOR MTG ONLY (6.10.2.13; 6.6.12)	
FLAT SURFACE PROVISION ONLY (6.10.2.14)	
MONITORS AND CABLES SUPPLIED BY (7.5.2.4)	
TEMPERATURE DETECTORS (7.5.2.3)	
TEMP. PROBES REQUIRED (7.5.2.3)	N / A
PROVISIONS FOR MOUNTING ONLY (6.10.2.2)	
RADIAL BEARING TEMP.	
NUMBER PER RADIAL BEARING	
THRUST BEARING TEMP.	
NUMBER PER THRUST BEARING ACTIVE SIDE	
NUMBER PER THRUST BEARING INACTIVE SIDE	
TEMP. GAUGES (WITH THERMOWELLS) (9.1.3.5)	
TEMP. GAUGE LOCATION	
SUPPLY UPPER/LOWER CASING RTD'S FOR WARMUP	
PRESSURE GAUGE TYPE	
PRESSURE GAUGE LOCATION	

REMARKS

PIPING & APPURTENANCES

TAG ALL ORIFICES (7.6.2.4)	*
SOCKET WELD UNION ON 1st SEAL GLAND NIPPLE (7.6.2.8)	*
MANIFOLD AUX PIPING SYSTEMS AT SKID EDGE (7.6.1.6)	*

SEAL SUPPORT SYSTEM MOUNTING

MOUNTED ON PUMP BASEPLATE (7.6.1.4)	YES
LOCATION ON OR OFF BASEPLATE (7.4.6) :	ON BASEPLATE
INTERCONNECTING PIPING BY	Supplier

MECHANICAL SEAL (6.8) (Note 7)

SEAL TYPE	API 682 MECHANICAL SEAL
API 682 DATASHEET ATTACHED :	YES
ADDITIONAL CENTRAL FLUSH PORT (6.8.9)	
HEATING JACKET REQ'D. (6.8.11)	
SEAL MANUFACTURER	*
SIZE / TYPE	* / *
MANUFACTURER CODE	*
ARRANGEMENT	DUAL
SEAL CHAMBER DATA:	
TEMPERATURE	* °C
PRESSURE	* kg/cm²g
FLOW	* m³/h
SEAL FLUSH PIPING PLAN:	11 + 52 (NOTE 11)
* in	*
SEAL FLUSH FLUID TYPE	*
PIPING ASSEMBLY	BY SUPPLIER
AUXILIARY PIPING PLAN:	*
* in	*
AUXILIARY FLUSH FLUID TYPE	*
PIPING ASSEMBLY	BY SUPPLIER

HEATING AND COOLING (6.1.23-6.1.27)

COOLING REQ'D		N / A
COOLING WATER PIPING PLAN (7.6.3.1)		
COOLING WATER PIPING FITTINGS		
COOLING WATER PIPING MATERIALS		
COOLING WATER REQUIREMENTS:	SUPPLY PRESSURE	kg/cm²g
	BEARING HOUSING	m³/h
	HEAT EXCHANGER	m³/h
	TOTAL COOLING WATER	m³/h
HEATING MEDIUM		N/A
HEATING PIPING		N/A

PRESSURE VESSEL DESIGN CODE REFERENCES

THESE REFERENCES SHALL BE PROVIDED BY THE MANUFACTURER

CASTING FACTORS USED IN DESIGN (PER TABLE 4)	
SOURCE OF MATERIAL PROPERTIES (6.3.5)	

WELDING AND REPAIRS (6.12.3.1)

THESE REFERENCES SHALL BE PROVIDED BY THE PURCHASER. (DEFAULT TO TABLE 11 IF NO PURCHASER PREFERENCE IS STATED)

ALTERNATIVE WELDING CODES AND STANDARDS	
WELDING REQUIREMENT (APPLICABLE CODE OR STANDARD)	DEFAULT PER TABLE 11
ALTERNATIVE WELDER/OPERATOR QUALIFICATION STANDARD	
ALTERNATIVE WELDING PROCEDURE QUALIFICATION STANDARD	
NON-PRESSURE RETAINING STRUCTURAL WELDING STANDARD (BASEPLATES OR SUPPORTS)	
STANDARD FOR MAGNETIC PARTICLE OR LIQUID PENETRANT EXAMINATION (PLATE EDGES)	
STANDARD FOR POSTWELD HEAT TREATMENT	
STANDARD FOR POSTWELD HEAT TREATMENT OF CASING FABRICATION WELDS	

MATERIAL INSPECTION

THESE REFERENCES SHALL BE PROVIDED BY THE PURCHASER

DEFAULT TO TABLE 14 **YES** TABLE 14 INSPECTION CLASS: **II**

ALTERNATIVE MATERIAL INSPECTIONS AND ACCEPTANCE CRITERIA (SEE TABLE 15, 8.2.2.5)

TYPE OF INSPECTION	METHOD	FOR FABRICATIONS	FOR CASTINGS
RADIOGRAPHY			
ULTRASONIC INSPECTION			
MAGNETIC PARTICLE INSPECTION			
LIQUID PENETRANT INSPECTION			
VISUAL INSPECTION (all surfaces)			

SURFACE PREPARATION, PAINT & SPARES

MANUFACTURER'S STANDARD (8.4.3.4)	NO
OTHER (SEE BELOW)	YES
SPECIFICATION NO.	Note 5
PUMP (Note 5):	
PUMP SURFACE PREPARATION	as per project specification
PRIMER	as per project specification
FINISH COAT	as per project specification
BASEPLATE (Note 5):	
SURFACE PREPARATION	as per project specification
PRIMER:	as per project specification
FINISH COAT	as per project specification
UNDERSIDE	as per project specification
SHIPMENT : (8.4)	EXPORT
EXPORT BOXING REQUIRED	YES
PREPARE FOR OUTDOOR STORAGE (NUMBER OF MONTHS) :	12
SPARE ROTOR ASSEMBLY PACKAGED FOR:	
ROTOR STORAGE ORIENTATION (9.2.8.2)	
SHIPPING & STORAGE CONTAINER SUITABLE FOR VERTICAL STORAGE (9.2.8.3) :	NO
N2 PURGE REQUIRED (9.2.8.4) :	
SPARE PARTS (10.3.4.2)	
START-UP	YES
NORMAL MAINTENANCE	

MASSES kg

ITEM No	PUMP	DRIVER	GEAR	BASE	TOTAL
P-007 A/B	*	*	*	*	*

OTHER PURCHASER REQUIREMENTS

COORDINATION MEETING REQUIRED	YES
MAXIMUM DISCHARGE PRESSURE TO INCLUDE (6.3.2)	
MAX RELATIVE DENSITY	
OPERATION TO TRIP SPEED	
MAX DIA. IMPELLERS AND/OR NO OF STAGES	
CONNECTION DESIGN APPROVAL (9.2.1.4)	
TORSIONAL ANALYSIS / REPORT (6.9.2)	
PROGRESS REPORTS (L.3.3)	YES
OUTLINE OF PROCEDURES USED FOR OPTIONAL TESTS (L.2.5)	
ADDITIONNAL DATA REQUIRING 20 YEARS RETENTION (8.2.1.1)	
LATERAL ANALYSIS REQUIRED (9.2.4.1.2)	YES
ROTOR DYNAMIC BALANCE TO 4W/N (6.9.3.5)	YES
VFD STEADY STATE FORCED RESPONSE ANALYSIS (6.9.2.3)	
TRANSIENT FORCED RESPONSE (6.9.2.4)	
BEARING LIFE CALCULATIONS REQUIRED (6.10.1.11)	YES
CASING RETIREMENT THICKNESS DRAWING [L.3.6.1 cc) 7) xiii)]	
CONNECTION BOLTING (7.6.1.7)	
VENDOR TO KEEP REPAIR AND HT RCDS (8.2.1.1.c)	YES
VENDOR SUBMIT TEST PROCEDURES (8.3.1.1)	YES
SUBMIT INSPECTION CHECK LIST (8.1.5)	YES
ACOUSTIC ANALYSIS OF CROSSOVER PASSAGE (BB3, BB5)(9.2.1.5)	
API-691 DOCUMENTATION REQUIRED (6.1.3.1)	

TEST

		Rev
SHOP INSPECTION (8.1.1.1)	YES	
PERF. CURVE & DATA APPROVAL REQ'D PRIOR TO SHIPMENT (8.3.3.4.6) :		
TEST WITH SUBSTITUTE SEAL (8.3.3.3.1)		
MATERIAL CERTIFICATION REQUIRED (6.12.1.8): **(N** CASING	YES	
IMPELLER	YES	
SHAFT	YES	
OTHER	As per API 610	
CASTING REPAIR WELD PROCEDURE APPROVAL REQD (6.12.2.5):	YES	
INSPECTION REQUIRED FOR CONNECTION WELDS (6.12.3.4)		
MAG PARTICLE		
RADIOGRAPHY		
LIQUID PENETRANT	YES	
ULTRASONIC		
HARDNESS TEST REQUIRED (6.12.1.14; 8.2.2.7)		
ADDITIONAL SUBSURFACE EXAMINATION (6.12.1.6) (8.2.1.3)		
FOR:	Casting	
METHOD:		
PMI TESTING REQUIRED (8.2.2.8)	YES	
COMPONENTS TO BE TESTED :	For Alloy Material	
RESIDUAL UNBALANCE TEST (J.4.1.2)		
NOTIFICATION OF SUCCESSFUL PERFORMANCE TEST (8.1.1.3) (8.3.3.4.6)		
BASEPLATE (NOZZLE LOAD) TEST (7.4.24)		
HYDROSTATIC TEST (8.3.2)	OBSERVE	
HYDROSTATIC TEST OF BOWLS & COLUMN (9.3.13.1)		
PERFORMANCE TEST (8.3.3)	WIT	
RETEST ON SEAL LEAKAGE (8.3.3.3.2)	OBSERVE	
TEST DATA POINTS TO (8.3.3.4)		
ALTERNATE TEST TOLERANCES PER (8.3.3.5)		
NPSH (8.3.4.3) (if margin < 1m (3ft)	WIT	
NPSH BASED ON 1ST STG ONLY ALLOWED (8.3.4.3.2)	N/A	
TEST NPSHA LIMITED TO 110% SITE NPSHA (8.3.3.7)		
RETEST REQUIRED AFTER FINAL HEAD ADJUSTMENT (8.3.3.8.2)		
COMPLETE UNIT TEST (8.3.4.4.1) **(Note 9)**	WIT	
SOUND LEVEL TEST (8.3.4.5)	WIT	
CLEANLINESS PRIOR TO FINAL ASSEMBLY (8.2.2.6)	NON-WIT	
LOCATION OF CLEANLINESS INSPECTION:		
CHECK FOR CO-PLANAR MOUNTING PAD FLATNESS (7.4.8)		
MECHANICAL RUN TEST UNTIL OIL TEMP STABLE (8.3.4.2.1)		
4 HR. MECH RUN AFTER OIL TEMP STABLE (8.3.4.2.2)	WIT	
BRG HSG RESONANCE TEST (8.3.4.7)		
STRUCTURAL RESONANCE TEST (9.3.9.2)		
REMOVE / INSPECT HYDRODYNAMIC BEARINGS AFTER TEST (9.2.7.4)		
AUXILIARY EQUIPMENT TEST (8.3.4.6)	NON-WIT	
EQUIPMENT TO BE INCLUDED IN AUXILLIARY TESTS :		
LOCATION OF AUXILIARY EQUIPMENT TEST :		
IMPACT TEST (6.12.4.3) PER EN 13445		
PER ASME SECTION VIII		
CASING DISASSEMBLY AFTER TEST (8.3.3.9)		
SPARE PARTS TEST (8.3.4.8)		

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NOTES

(*) = VTA (VENDOR to advise and shall fill detail value or data before Purchase Order)

1 Margin NPSH shall be minimum 1 meter (3 ft) otherwise Witness NPSH Test is required.

2 Material certificates shall be provided for 3.1 for wetted parts & pressure parts and 2.2 for others as per ISO 10474 Standard.

3 The Pump shall be designed as per below requirements:
- TMBP-SP001-ME-DBS-D101 — Mechanical Design Basis
- TMBP-SP001-PR-DST-D108 —Process Datasheet – KOD Pump P-006 A/B
- TMBP-SP001-PR-UFD-D103_UTILITY_FLOW_DIAGRAM_FOR_FLARE_SYSTEM
- TMBP-SP001-ME-SPE-D103 — Specification – API Centrifugal Pump
- API 610 Latest Edition — Centrifugal Pumps for Petroleum, Petrochemical and Natural Gas Industries
- API 682 — Pumps – Shaft Sealing Systems for Centrifugal and Rotary Pumps

4 Motor shall be designed as per below requirements:
- Motor shall be as per IEC Standard & shall be sized for End of Curve (EOC) operation
- TMBP-SP001-EL-DST-D102 — Datasheet for Induction Motor
- TMBP-SP001-EL-SPE-D106 — Specification for Induction Motor

5 Finish/top color shall comply as per Document Painting & Coating TMBP-SP001-PI-SPE-D109 “Piping Painting & Lettering Specification”.

6 Drain & Vent connection shall be provided with isolation valve and blind flange.

7 Mechanical seal shall be designed as per below requirement:
- Mechanical seal shall be compatible with process liquid
- VENDOR shall check, design and guarantee that material of mechanical seal is suitable as per process liquid
- VENDOR shall provide mechanical seal selection from seal manufacturer before Purchase Order (P.O)

8 VENDOR shall check and guarantee that material of the pump is suitable as per process liquid before Purchase Order (P.O).

9 Complete Unit Test of the pump shall include with job motor & job mechanical seal.

10 Pump should be completed with oil level glass/constant level oiler.

11 Pump actual capacity shall be ±10% from design capacity.

12 Pump operation shall be 2×100%, 1 (one) running and 1 (one) standby.

13 VENDOR shall recalculate actual power rating before Purchase Order (PO).

14 Temperature of oil at bearing chamber shall not exceed 140 °F and max 180 °F will shut down the pump.

15 VENDOR shall include below requirement:
- Factory Acceptance Test (FAT) execution
- Motor solo run, site realignment, SAT including commissioning supervision
- Certificate material 3.1 and 3.2 for critical part shaft and impeller
- First Filling: Lube oil/Grease
- Mating Flange, Counter Flange (if required)
- Insulation for bolt connection at flange suction and discharge

GENERAL

APPLICABLE TO:	PROPOSAL	APPLICABLE NATIONAL / INTERNATIONAL STANDARD :	API-610 (Note 3)	Rev
CLIENT	Pertamina EP Regional 1 Zona 4	UNIT		
SITE	Fasilitas Produksi Stasiun Pengumpul (SP)	SERVICE	Closed Drain Pump	
NO. REQ	2x100% (Note 12)	PUMP SIZE	*	TYPE: OH2 — No. STAGES: 1 (One)
MANUFACTURER	*	MODEL	*	SERIAL NO. *
EQUIPMENT NUMBER	P-007 A/B			

LIQUID CHARACTERISTICS

	Units						
LIQUID TYPE OR NAME :		Mixed Oil & Water					
		Rated	Normal	MAX	MIN		
SERVICE :		Process Liquid					
VAPOR PRESSURE :	(psi a)	2.75					
SG @ Pump. Temp (3.1.53) :		0.88					
SPECIFIC HEAT :	(Btu/lb)						
VISCOSITY @ Pump. Temp. :	(cP)	12.66					

OPERATING CONDITIONS (6.1.3)

	Units	Rated	Normal	MAX	MIN	Alt Condition 1 {Name 3}	Alt Condition 2 {Name 4}
NPSHA Datum:		C.L. Impeller					
PUMPING TEMPERATURE :	(ºF)	65.00					
FLOW (Note 11) :	(BPD)	550					
DISCHARGE PRESSURE (6.3.2) :	(psig)	30.0					
SUCTION PRESSURE :	(psig)	1.65					
DIFFERENTIAL PRESSURE :	(psi)	28.35					
DIFFERENTIAL HEAD :	(ft)	74.25					
NPSHA (Note 1) :	(ft)	35.81					
HYDRAULIC POWER:	(hp)	0.30					

SERVICE (CONTINUOUS/INTERMITTENT) :	INTERMITTENT	INSTALLATION LOCATION :	OUTDOOR
• IF INTERMITTENT, NO. OF STARTS :		IF INDOOR, TEMPERATURE: MAX: ºC	MIN: ºC
PUMPS OPERATE IN:	ALONE	ELECTRIC AREA CLASSIFICATION (6.1.29):	HAZARDOUS
CORROSION DUE TO : (6.12.1.9)	N/A	CLASS	ZONE 2
EROSION DUE TO : (6.12.1.9)		GROUP IIA	LIQUID CLASS: FLAMMABLE
H2S CONCENTRATION (ppm) : (6.12.1.13)	N/A	TEMP CLASS T3	
CHLORIDE CONCENTRATION	N/A	PUMP VALVE START POSITION :	Open Valve (Fully Loaded)
PARTICULATE SIZE (DIA IN MICRONS)	Note 1		
PARTICULATE CONCENTRATION (ppm)	N/A		
CORROSION ALLOWANCE (6.3.10) :			
DESIGN NOTES:	Tropicalization Required		

PERFORMANCE

PROPOSAL CURVE NO.	*	RPM	*	
IMPELLER DIA.: RATED	*	MAX. *	MIN. *	(in)
RATED POWER :	* HP	EFFICIENCY	*	%
RATED CURVE BEP FLOW (at rated impeller diameter) (6.1.16) :			*	(bpd)
MIN FLOW : THERMAL	* (bpd)	STABLE	*	(bpd)
PREFERRED OPERATING REGION (6.1.16)	*	to	*	(bpd)
ALLOWABLE OPERATING REGION (6.9.4.1)	*	to	*	(bpd)
MAX HEAD @ RATED IMPELLER			*	(ft)
MAX POWER @ RATED IMPELLER			*	(HP)
NPSH3 AT RATED FLOW (6.1.9) :			*	(ft)
CENTERLINE OF PUMP TO NPSHA DATUM			*	(ft)
NPSH MARGIN AT RATED FLOW (6.1.10) :			*	(ft)
SPECIFIC SPEED (6.1.17)		m3/h, rpm, m	*	(m3/h, rpm, m)
SUCTION SPECIFIC SPEED LIMIT (6.1.11)		m3/h, rpm, m	*	
SUCTION SPECIFIC SPEED (6.1.11)		m3/h, rpm, m	*	(m3/h, rpm, m)
MAX. ALLOWABLE SOUND PRESSURE LEVEL (6.1.19)			85 @1 m	(dBA)
ESTIMATED MAX. SOUND PRESSURE LEVEL			*	(dBA)
MAX. ALLOWABLE SOUND POWER LEVEL (6.1.19)				(dB)
ESTIMATED MAX SOUND POWER LEVEL			*	(dB)

DRIVER (7.1) (Note 4)

DRIVER TYPE	Electric Motor
GEAR	NO
VARIABLE SPEED REQUIRED	NO
SOURCE OF VARIABLE SPEED	N/A
MANUFACTURER	*
NAMEPLATE POWER : HP	*
NOMINAL RPM	*
RATED LOAD RPM	*
FRAME	*
ORIENTATION	HORIZONTAL
LUBRICATION	*
BEARING TYPE	Anti Friction
RADIAL	*
THRUST	*
STARTING METHOD	DOL
DRIVER DATASHEET NUMBER	TMBP-SP001-EL-DST-D102
DRIVE DESIGN STANDARD	IEC

CONSTRUCTION

API PUMP TYPE (Table 1): OH2

NOZZLE CONNECTIONS (6.4.2):

	Size	Facing	Rating	Position
SUCTION	3" (*)	RF	300	END
DISCHARGE	2" (*)	RF	300	TOP

PRESSURE CASING AUX. CONNECTIONS: (6.4.3)

	No.	Size	Type	Facing	Rating	Posn.
BALANCE/LEAK OFF						
DRAIN **(Note 6)**		(*)		RF	(*)	
VENT **(Note 6)**		(*)		RF	(*)	
PRESSURE GAGE						
TEMP GAGE						
WARM-UP LINE						

DRAIN VALVE SUPPLIED BY (7.6.2.5)	SUPPLIER
DRAINS MANIFOLDED (7.6.2.5)	NO
VENT VALVE SUPPLIED BY	SUPPLIER
VENTS MANIFOLDED	NO
THREADED CONS FOR PIPELINE SERVICE & < 55°C (6.4.3.2)	
SPECIAL FITTINGS FOR TRANSITIONING (6.4.3.3)	
CYLINDRICAL THREADS REQUIRED (6.4.3.2)	
MACHINED AND STUDDED AUX CONNECTIONS (6.4.3.6)	
ROUTE DRAIN TO SKID EDGE	YES

CASING MOUNTING (6.3.14, 9.3.8.3): CENTERLINE

CASING TYPE: SINGLE VOLUTE

OH3 BACK-PULLOUT LIFTING DEVICE REQD. (9.1.2.6): N/A

CASING PRESSURE RATING:

MAWP (6.3.6) : * psig @ * (ºF)

HYDROTEST (8.3.2.2) : * psig @ AMB (ºF)

WETTING AGENT REQUIRED FOR HYDROTEST (8.3.2.7) :	*
HYDROTEST OH PUMP AS ASSEMBLY (8.3.2.14) :	*
SUCT'N PRESS. REGIONS DESIGNED FOR MAWP (6.3.8) :	YES

ROTOR:

SHAFT FLEXIBILITY INDEX (SFI) (9.1.1.3)	*
FIRST CRITICAL SPEED WET (MULTISTAGE PUMPS ONLY)	N / A
COMPONENT BALANCE TO ISO 1940-1, G1 (6.9.3.4) :	YES
SHRINK-FIT-LIMITED MOVEMENT IMPELLERS (9.2.2.3)	*
ROTATION (VIEWED FROM COUPLING END) :	CW
IMPELLERS INDIVIDUALLY SECURED (6.6.3) :	YES

COUPLING: (7.2)

MANUFACTURER	*
MODEL	Flexible Disc Type with Spacer (*)
RATING (kW / 100 RPM)	*
SPACER LENGTH (7.2.2.d)	* mm
SERVICE FACTOR (7.2.3)	1.5
RIGID-TYPE	NO
COUPLING WITH HYDRAULIC FIT (7.2.9)	*
COUPLING WITH PROPRIETARY CLAMPING DEVICE (7.2.10)	*
COUPLING IN COMPLIANCE WITH (7.2.4)(7.2.2.f)	API 671

GUARDS (7.3)

COUPLING AND SHAFT GUARD STANDARD (7.3)	*
IGNITION HAZARD ASSESSMENT PER EN 13463-1 (7.3.2.2; 7.3.3.4)	*
COUPLING GUARD MATERIAL (7.3.2.1; 7.3.3.3) :	Non-sparking
SHAFT GUARD MATERIAL (7.3.2.1; 7.3.3.3) :	*
SPARK RESISTANT MATERIAL REQUIRED (7.3.2.1) :	YES

MATERIAL (6.12.1.1) (Note 8)

ANNEX H CLASS	S-6	
MINIMUM DESIGN METAL TEMPERATURE (6.12.4.1)	0	°C
MAXIMUM ALLOWABLE TEMPERATURE (3.1.19)		°C
REDUCED-HARDNESS MATERIALS REQ'D (6.12.1.14)		
APPLICABLE HARDNESS STANDARD (6.12.1.14)		
BARREL :		
CASE :	CS	
DIFFUSERS		
IMPELLER :	12% Hardened	
IMPELLER WEAR RING :	12% Hardened	
CASE WEAR RING :	12% Hardened	
SHAFT:	4140 Alloy Steel	
BOWL (IF VS-TYPE)		

BEARINGS AND BEARING HOUSINGS (6.10.1)

BEARING (TYPE / NUMBER):

RADIAL	*	/	*
THRUST	*	/	*

REVIEW AND APPROVE THRUST BEARING SIZE : (9.2.5.2.6)	
LUBRICATION (6.11.3) (9.3.12.4) : **(Note 11, 14)**	FLOOD
PRESSURE LUBE SYSTEM TO API-614, CHAPTER :	*
API 614 DATASHEETS ATTACHED (9.2.6.4)	*
PRESSURIZED LUBE OIL SYSTEM MOUNTED ON BASEPLATE:	*
LOCATION OF PRESSURIZED LUBE OIL SYSTEM:	*
INTERCONNECTING PIPING PROVIDED BY	*
OIL VISC. ISO GRADE	*
VENT-TO-HOUSING CONSTANT LEVEL OILER (6.10.2.4) :	REQUIRED
OIL MIST PROVISIONS (6.11.3)	N/A
GREASE LUBRICATION (6.11.4)	N/A

BASEPLATE

API BASEPLATE NUMBER (ANNEX D) :	*
IF NON-STD BASEPLATE DIMENSIONS (LxW) (	
BASEPLATE CONSTRUCTION (7.4)	*
BASEPLATE DRAINAGE (7.4.1)	*
MOUNTING :	GROUTED
NON-GROUT CONSTRUCTION (7.4.1.e):	
SUPPLIED WITH : ● GROUT AND VENT HOLES	YES
● DRAIN CONNECTION	YES
DEMONSTRATE BASEPLATE PAD FLATNESS (7.4.9)	*
PROVIDE SPACER PLATE UNDER ALL EQUIPMENT FEET (7.4.10)	
BOLT OH 3/4/5 PUMP TO PAD / FOUNDATION :	N / A
PROVIDE SOLEPLATE FOR OH 3/4/5 PUMPS (9.1.2) :	N / A

REMARKS:

DISCHARGE CIRCUIT PIPING DESIGN:

DESIGN PRESSURE _____ psig

DESIGN TEMPERATURE _____ ºF

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INSTRUMENTATION

INSTRUMENTATION PER API-670 (7.5.2)	N / A
ACCELEROMETERS (7.5.2.1)	
NUMBER OF ACCELEROMETERS	N/A
MOUNTING LOCATION:	
PROVISION FOR MTG ONLY (6.10.2.13)	
FLAT SURFACE REQUIRED (6.10.2.14)	
VIBRATION PROBES (7.5.2.2)	
VIBRATION PROBES REQUIRED (7.5.2.2)	N / A
NUMBER PER RADIAL BEARING	
NUMBER PER AXIAL BEARING	
THREADED PROVISION FOR MTG ONLY (6.10.2.13; 6.6.12)	
FLAT SURFACE PROVISION ONLY (6.10.2.14)	
MONITORS AND CABLES SUPPLIED BY (7.5.2.4)	
TEMPERATURE DETECTORS (7.5.2.3)	
TEMP. PROBES REQUIRED (7.5.2.3)	N / A
PROVISIONS FOR MOUNTING ONLY (6.10.2.2)	
RADIAL BEARING TEMP.	
NUMBER PER RADIAL BEARING	
THRUST BEARING TEMP.	
NUMBER PER THRUST BEARING ACTIVE SIDE	
NUMBER PER THRUST BEARING INACTIVE SIDE	
TEMP. GAUGES (WITH THERMOWELLS) (9.1.3.5)	
TEMP. GAUGE LOCATION	
SUPPLY UPPER/LOWER CASING RTD'S FOR WARMUP	
PRESSURE GAUGE TYPE	
PRESSURE GAUGE LOCATION	

REMARKS

PIPING & APPURTENANCES

TAG ALL ORIFICES (7.6.2.4)	*
SOCKET WELD UNION ON 1st SEAL GLAND NIPPLE (7.6.2.8)	*
MANIFOLD AUX PIPING SYSTEMS AT SKID EDGE (7.6.1.6)	*

SEAL SUPPORT SYSTEM MOUNTING

MOUNTED ON PUMP BASEPLATE (7.6.1.4)	YES
LOCATION ON OR OFF BASEPLATE (7.4.6) :	ON BASEPLATE
INTERCONNECTING PIPING BY	Supplier

MECHANICAL SEAL (6.8) (Note 7)

SEAL TYPE	API 682 MECHANICAL SEAL
API 682 DATASHEET ATTACHED :	YES
ADDITIONAL CENTRAL FLUSH PORT (6.8.9)	
HEATING JACKET REQ'D. (6.8.11)	
SEAL MANUFACTURER	*
SIZE / TYPE	* / *
MANUFACTURER CODE	*
ARRANGEMENT	DUAL
SEAL CHAMBER DATA:	
TEMPERATURE	* °C
PRESSURE	* kg/cm²g
FLOW	* m³/h
SEAL FLUSH PIPING PLAN:	11 + 52 (NOTE 11)
* in	*
SEAL FLUSH FLUID TYPE	*
PIPING ASSEMBLY	BY SUPPLIER
AUXILIARY PIPING PLAN:	*
* in	*
AUXILIARY FLUSH FLUID TYPE	*
PIPING ASSEMBLY	BY SUPPLIER

HEATING AND COOLING (6.1.23-6.1.27)

COOLING REQ'D		N / A
COOLING WATER PIPING PLAN (7.6.3.1)		
COOLING WATER PIPING FITTINGS		
COOLING WATER PIPING MATERIALS		
COOLING WATER REQUIREMENTS:	SUPPLY PRESSURE	kg/cm²g
	BEARING HOUSING	m³/h
	HEAT EXCHANGER	m³/h
	TOTAL COOLING WATER	m³/h
HEATING MEDIUM		N/A
HEATING PIPING		N/A

PRESSURE VESSEL DESIGN CODE REFERENCES

THESE REFERENCES SHALL BE PROVIDED BY THE MANUFACTURER

CASTING FACTORS USED IN DESIGN (PER TABLE 4)	
SOURCE OF MATERIAL PROPERTIES (6.3.5)	

WELDING AND REPAIRS (6.12.3.1)

THESE REFERENCES SHALL BE PROVIDED BY THE PURCHASER. (DEFAULT TO TABLE 11 IF NO PURCHASER PREFERENCE IS STATED)

ALTERNATIVE WELDING CODES AND STANDARDS	
WELDING REQUIREMENT (APPLICABLE CODE OR STANDARD)	DEFAULT PER TABLE 11
ALTERNATIVE WELDER/OPERATOR QUALIFICATION STANDARD	
ALTERNATIVE WELDING PROCEDURE QUALIFICATION STANDARD	
NON-PRESSURE RETAINING STRUCTURAL WELDING STANDARD (BASEPLATES OR SUPPORTS)	
STANDARD FOR MAGNETIC PARTICLE OR LIQUID PENETRANT EXAMINATION (PLATE EDGES)	
STANDARD FOR POSTWELD HEAT TREATMENT	
STANDARD FOR POSTWELD HEAT TREATMENT OF CASING FABRICATION WELDS	

MATERIAL INSPECTION

THESE REFERENCES SHALL BE PROVIDED BY THE PURCHASER

DEFAULT TO TABLE 14 **YES** TABLE 14 INSPECTION CLASS: **II**

ALTERNATIVE MATERIAL INSPECTIONS AND ACCEPTANCE CRITERIA (SEE TABLE 15, 8.2.2.5)

TYPE OF INSPECTION	METHOD	FOR FABRICATIONS	FOR CASTINGS
RADIOGRAPHY			
ULTRASONIC INSPECTION			
MAGNETIC PARTICLE INSPECTION			
LIQUID PENETRANT INSPECTION			
VISUAL INSPECTION (all surfaces)			

SURFACE PREPARATION, PAINT & SPARES

MANUFACTURER'S STANDARD (8.4.3.4)	NO
OTHER (SEE BELOW)	YES
SPECIFICATION NO.	Note 5
PUMP (Note 5):	
PUMP SURFACE PREPARATION	as per project specification
PRIMER	as per project specification
FINISH COAT	as per project specification
BASEPLATE (Note 5):	
SURFACE PREPARATION	as per project specification
PRIMER:	as per project specification
FINISH COAT	as per project specification
UNDERSIDE	as per project specification
SHIPMENT : (8.4)	EXPORT
EXPORT BOXING REQUIRED	YES
PREPARE FOR OUTDOOR STORAGE (NUMBER OF MONTHS) :	12
SPARE ROTOR ASSEMBLY PACKAGED FOR:	
ROTOR STORAGE ORIENTATION (9.2.8.2)	
SHIPPING & STORAGE CONTAINER SUITABLE FOR VERTICAL STORAGE (9.2.8.3) :	NO
N2 PURGE REQUIRED (9.2.8.4) :	
SPARE PARTS (10.3.4.2)	
START-UP	YES
NORMAL MAINTENANCE	

MASSES kg

ITEM No	PUMP	DRIVER	GEAR	BASE	TOTAL
P-007 A/B	*	*	*	*	*

OTHER PURCHASER REQUIREMENTS

COORDINATION MEETING REQUIRED	YES
MAXIMUM DISCHARGE PRESSURE TO INCLUDE (6.3.2)	
MAX RELATIVE DENSITY	
OPERATION TO TRIP SPEED	
MAX DIA. IMPELLERS AND/OR NO OF STAGES	
CONNECTION DESIGN APPROVAL (9.2.1.4)	
TORSIONAL ANALYSIS / REPORT (6.9.2)	
PROGRESS REPORTS (L.3.3)	YES
OUTLINE OF PROCEDURES USED FOR OPTIONAL TESTS (L.2.5)	
ADDITIONNAL DATA REQUIRING 20 YEARS RETENTION (8.2.1.1)	
LATERAL ANALYSIS REQUIRED (9.2.4.1.2)	YES
ROTOR DYNAMIC BALANCE TO 4W/N (6.9.3.5)	YES
VFD STEADY STATE FORCED RESPONSE ANALYSIS (6.9.2.3)	
TRANSIENT FORCED RESPONSE (6.9.2.4)	
BEARING LIFE CALCULATIONS REQUIRED (6.10.1.11)	YES
CASING RETIREMENT THICKNESS DRAWING [L.3.6.1 cc) 7) xiii)]	
CONNECTION BOLTING (7.6.1.7)	
VENDOR TO KEEP REPAIR AND HT RCDS (8.2.1.1.c)	YES
VENDOR SUBMIT TEST PROCEDURES (8.3.1.1)	YES
SUBMIT INSPECTION CHECK LIST (8.1.5)	YES
ACOUSTIC ANALYSIS OF CROSSOVER PASSAGE (BB3, BB5)(9.2.1.5)	
API-691 DOCUMENTATION REQUIRED (6.1.3.1)	

TEST

		Rev
SHOP INSPECTION (8.1.1.1)	YES	
PERF. CURVE & DATA APPROVAL REQ'D PRIOR TO SHIPMENT (8.3.3.4.6) :		
TEST WITH SUBSTITUTE SEAL (8.3.3.3.1)		
MATERIAL CERTIFICATION REQUIRED (6.12.1.8): **(N** CASING	YES	
IMPELLER	YES	
SHAFT	YES	
OTHER	As per API 610	
CASTING REPAIR WELD PROCEDURE APPROVAL REQD (6.12.2.5):	YES	
INSPECTION REQUIRED FOR CONNECTION WELDS (6.12.3.4)		
MAG PARTICLE		
RADIOGRAPHY		
LIQUID PENETRANT	YES	
ULTRASONIC		
HARDNESS TEST REQUIRED (6.12.1.14; 8.2.2.7)		
ADDITIONAL SUBSURFACE EXAMINATION (6.12.1.6) (8.2.1.3)		
FOR:	Casting	
METHOD:		
PMI TESTING REQUIRED (8.2.2.8)	YES	
COMPONENTS TO BE TESTED :	For Alloy Material	
RESIDUAL UNBALANCE TEST (J.4.1.2)		
NOTIFICATION OF SUCCESSFUL PERFORMANCE TEST (8.1.1.3) (8.3.3.4.6)		
BASEPLATE (NOZZLE LOAD) TEST (7.4.24)		
HYDROSTATIC TEST (8.3.2)	OBSERVE	
HYDROSTATIC TEST OF BOWLS & COLUMN (9.3.13.1)		
PERFORMANCE TEST (8.3.3)	WIT	
RETEST ON SEAL LEAKAGE (8.3.3.3.2)	OBSERVE	
TEST DATA POINTS TO (8.3.3.4)		
ALTERNATE TEST TOLERANCES PER (8.3.3.5)		
NPSH (8.3.4.3) (if margin < 1m (3ft)	WIT	
NPSH BASED ON 1ST STG ONLY ALLOWED (8.3.4.3.2)	N/A	
TEST NPSHA LIMITED TO 110% SITE NPSHA (8.3.3.7)		
RETEST REQUIRED AFTER FINAL HEAD ADJUSTMENT (8.3.3.8.2)		
COMPLETE UNIT TEST (8.3.4.4.1) **(Note 9)**	WIT	
SOUND LEVEL TEST (8.3.4.5)	WIT	
CLEANLINESS PRIOR TO FINAL ASSEMBLY (8.2.2.6)	NON-WIT	
LOCATION OF CLEANLINESS INSPECTION:		
CHECK FOR CO-PLANAR MOUNTING PAD FLATNESS (7.4.8)		
MECHANICAL RUN TEST UNTIL OIL TEMP STABLE (8.3.4.2.1)		
4 HR. MECH RUN AFTER OIL TEMP STABLE (8.3.4.2.2)	WIT	
BRG HSG RESONANCE TEST (8.3.4.7)		
STRUCTURAL RESONANCE TEST (9.3.9.2)		
REMOVE / INSPECT HYDRODYNAMIC BEARINGS AFTER TEST (9.2.7.4)		
AUXILIARY EQUIPMENT TEST (8.3.4.6)	NON-WIT	
EQUIPMENT TO BE INCLUDED IN AUXILLIARY TESTS :		
LOCATION OF AUXILIARY EQUIPMENT TEST :		
IMPACT TEST (6.12.4.3) PER EN 13445		
PER ASME SECTION VIII		
CASING DISASSEMBLY AFTER TEST (8.3.3.9)		
SPARE PARTS TEST (8.3.4.8)		

Rev.

NOTES

(*) = VTA (VENDOR to advise and shall fill detail value or data before Purchase Order)

1 Margin NPSH shall be minimum 1 meter (3 ft) otherwise Witness NPSH Test is required.

2 Material certificates shall be provided for 3.1 for wetted parts & pressure parts and 2.2 for others as per ISO 10474 Standard.

3 The Pump shall be designed as per below requirements:
- TMBP-SP001-ME-DBS-D101 — Mechanical Design Basis
- TMBP-SP001-PR-CAL-D112 — Process Calculation Closed Drain Pump (P-007 A/B)
- TMBP-SP001-PR-DST-D109 —Process Datasheet - Closed Drain Pump P-007 A/B
- TMBP-01-PR-PID-116 —Piping and Instrument diagram for Closed Drain Vessel (V-007)
- TMBP-SP001-ME-SPE-D103 — Specification – API Centrifugal Pump
- API 610 Latest Edition — Centrifugal Pumps for Petroleum, Petrochemical and Natural Gas Industries
- API 682 — Pumps – Shaft Sealing Systems for Centrifugal and Rotary Pumps

4 Motor shall be designed as per below requirements:
- Motor shall be as per IEC Standard & shall be sized for End of Curve (EOC) operation
- TMBP-SP001-EL-DST-D102 — Datasheet for Induction Motor
- TMBP-SP001-EL-SPE-D106 — Specification for Induction Motor

5 Finish/top color shall comply as per Document Painting & Coating TMBP-SP001-PI-SPE-D109 “Piping Painting & Lettering Specification”.

6 Drain & Vent connection shall be provided with isolation valve and blind flange.

7 Mechanical seal shall be designed as per below requirement:
- Mechanical seal shall be compatible with process liquid
- VENDOR shall check, design and guarantee that material of mechanical seal is suitable as per process liquid
- VENDOR shall provide mechanical seal selection from seal manufacturer before Purchase Order (P.O)

8 VENDOR shall check and guarantee that material of the pump is suitable as per process liquid before Purchase Order (P.O).

9 Complete Unit Test of the pump shall include with job motor & job mechanical seal.

10 Pump should be completed with oil level glass/constant level oiler.

11 Pump actual capacity shall be ±10% from design capacity.

12 Pump operation shall be 2×100%, 1 (one) running and 1 (one) standby.

13 VENDOR shall recalculate actual power rating before Purchase Order (PO).

14 Temperature of oil at bearing chamber shall not exceed 140 °F and max 180 °F will shut down the pump.

15 VENDOR shall include below requirement:
- Factory Acceptance Test (FAT) execution
- Motor solo run, site realignment, SAT including commissioning supervision
- Certificate material 3.1 and 3.2 for critical part shaft and impeller
- First Filling: Lube oil/Grease
- Mating Flange, Counter Flange (if required)
- Insulation for bolt connection at flange suction and discharge